Evaluating Deployed-Selection Robustness in Forecasting Benchmarks

Anonymous Authors[1]

Abstract

Forecasting benchmarks commonly rank models by forecast-side metrics, and these metrics remain the appropriate object when the target is predictive quality. In deployment-facing forecasting settings, however, forecasts may be consumed through a fixed forecast-to-decision interface, such as a threshold alert, hysteresis threshold, or top-k budget alert, before deployed-side scoring. We do not propose a new forecaster, training objective, or universal deployed metric; instead, we study whether forecast-side ranking preserves deployed-side model selection under the same fixed interface. Across a synthetic zero-friction anchor, forecasting-native event-micro and Traffic-Hourly tasks, and inventory corroboration, we find that forecast-side winners can become recurrently deployed-suboptimal in moderate-to-high friction regimes in the studied settings. The implication is not to invalidate forecast-side evaluation for prediction, but to report fixed-interface deployed-selection robustness alongside forecast-side metrics when forecasting benchmarks are used for deployment-facing model selection.

1. Introduction

Forecasting benchmarks typically compare models using forecast-side metrics, and those metrics are the right object when the benchmark asks about predictive quality. Some deployment-facing forecasting evaluations, however, consume forecasts through a fixed forecast-to-decision interface before deployed-side scoring. By a fixed interface we mean a common forecast-to-action rule applied to every forecaster before scoring, such as triggering an alert when $p_t > \tau$, changing state only if the score moves by at least δ, or issuing the top-k alerts under a fixed budget. When that interface includes switching or deployment friction, forecast-side ranking can diverge from deployed-side model selection: the model that wins on the forecast metric need not remain the model that performs best after the fixed interface is applied. This is not an argument for replacing forecast benchmarks with a single deployed metric. Because deployed metrics depend on a specific interface, simulator, and friction model, they are not a drop-in replacement for general forecasting benchmarks. Our claim is narrower: deployment-facing benchmarks should report whether forecast-side ranking preserves deployed-side model selection under the chosen fixed interface.

This paper frames the mismatch as a reporting diagnostic for deployment-facing forecasting benchmarks. We call the diagnostic fixed-interface deployed-selection robustness: whether the forecast-side winner remains the deployed-side winner after forecasts are passed through the same fixed interface. Our main empirical question, Q2, asks whether this preservation holds when the interface is fixed and the forecaster varies. Q1 is retained only as mechanism support: it holds the proposed action path fixed and asks how frictional execution can separate proposed and realized actions. This framing keeps forecast-side metrics as prediction measures while asking what deployment-facing evaluation should report alongside them.

This paper makes three narrow contributions.

- First, we define fixed-interface deployed-selection robustness as a reporting diagnostic: whether a forecast-side winner remains the deployed-side winner after forecasts are passed through the same fixed interface.
- Second, we show that, under the studied fixed interfaces, forecast-side winners can become recurrently deployed-suboptimal as friction increases, especially in moderate-to-high friction regimes.
- Third, we organize evidence in a Q2-first structure, with event-micro and Traffic-Hourly as main forecasting-native evidence, inventory as operational corroboration, and Q1 only as mechanism support.

We use Q1/Q2 only to separate mechanism from model-selection evidence. Q1 fixes the proposed action path and varies the interface as an exact-control check. Q2 fixes the interface and varies the forecaster as the main model-selection question.

[1]Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

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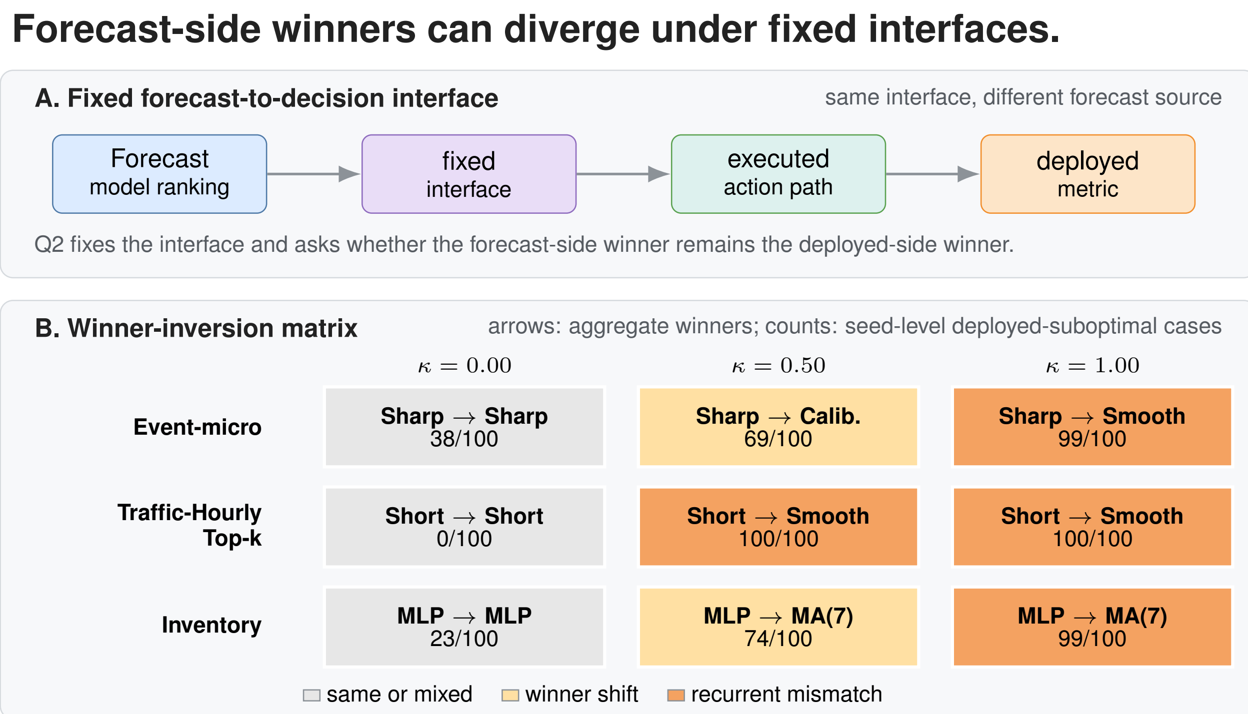

Figure 1. Fixed-interface evaluation can change model selection. Panel A shows the Q2 setting: the interface is fixed, the forecaster varies, and the question is whether the forecast-side winner remains the deployed-side winner. Panel B reports aggregate forecast-side and deployed-side winners across domains and friction levels. Cell counts denote seed-level deployed-suboptimal cases for the forecast-selected model; darker orange marks recurrent mismatch.

2. Results

2.1. Q2: Fixed interface and deployed-selection robustness

Q2 asks the paper's headline reporting question: under one fixed forecast-to-decision interface, does the forecast-side winner remain the deployed-side winner? Table 1 reports the resulting deployed-selection robustness report card. The zero-friction synthetic row is a sanity anchor. Event-micro and Traffic-Hourly provide forecasting-native Q2 evidence under two different fixed-interface structures, while inventory is included only as operational corroboration. Figure 1 visualizes the main winner-inversion evidence across the forecasting-native and corroboration rows.

In the report card, Event-micro shows the clearest primary Q2 pattern: the forecast-side winner remains Reactive sharp, but the deployed winner changes to Calibrated baseline at friction 0.50 and Lagged smoother at friction 1.00. The deployed-selection consequence is recurrent: the forecast-selected model is deployed-suboptimal in 69/100 and 99/100 seeds at these two friction levels. The event-micro result is not confined to the canonical threshold or Brier ranking: at frictions 0.50/1.00, deployed-suboptimality remains 77/100 and 98/100 under an alternate threshold, 74/100 and 99/100 under log-loss reranking, and 55/100 and 82/100 under a hysteresis interface, where the direction is preserved with smaller magnitude.

A second forecasting-native Traffic-Hourly Top-k alert task repeats the same selection failure under a fixed-budget alert interface. In the selected Top-k setting ($k = 249$), Reactive short is both the forecast-side winner and the deployed winner at zero friction, with exact agreement 1.0. Once switching costs are introduced, however, the forecast-side winner remains Reactive short while the deployed winner shifts to Lagged smoother, and deployed-suboptimality reaches $100/100$ seeds at both friction 0.5 and friction 1.0. A relative-rank Traffic variant shows the same qualitative pattern in the appendix.

Inventory is used only as operational corroboration, not as main forecasting-native evidence. At friction 0.50 and 1.00, the forecast-selected Small MLP is deployed-suboptimal in 74/100 and 99/100 seeds, respectively, under the fixed replenishment interface; the report card includes only the high-friction inventory row to keep this role bounded.

Table 1. Deployed-selection robustness report card. Each row reports one domain × fixed interface × friction cell and records whether the forecast-side winner remains the deployed-side winner after the same fixed interface is applied. Agreement rate is the fraction of matched seeds for which the forecast-side selected model is also the deployed-side best model; deployed-suboptimal share is the complementary fraction; deployed gap is the paired deployed-utility shortfall of the forecast-selected model relative to the deployed winner. Subopt. reports the seed-level deployed-suboptimal share/count. The 95% CI column reports the paired-bootstrap CI for the mean deployed gap, not the binomial CI for the suboptimal share.

Domain	Interface	κ	Forecast winner	Deployed winner	Agree.	Subopt.	Mean gap	95% CI	Role
Synthetic	zero-friction anchor	0.00	Naive last	Naive last	1.00	0.00 (0/20)	0.000	[0.000, 0.000]	Sanity
Event-micro	threshold τ=0.55	0.50	Reactive sharp	Calibrated	0.31	0.69 (69/100)	0.011	[0.009, 0.014]	Primary
Event-micro	threshold τ=0.55	1.00	Reactive sharp	Smoother	0.01	0.99 (99/100)	0.057	[0.052, 0.063]	Primary
Traffic Top-k	budget k=249	0.50	Reactive short	Smoother	0.00	1.00 (100/100)	7.016	[6.756, 7.283]	Second
Traffic Top-k	budget k=249	1.00	Reactive short	Smoother	0.00	1.00 (100/100)	24.651	[24.226, 25.079]	Second
Inventory	replenishment	1.00	Small MLP	MA(7)	0.01	0.99 (99/100)	1.029	[0.946, 1.111]	Corrob.

Q1 mechanism support. Q1 explains why divergence can arise; Q2 is the actual benchmark-selection claim. Holding the proposed action path fixed, synthetic and inventory show that frictional interfaces can separate proposed and realized actions. Synthetic yields an increasing proposal-versus-realization gap away from zero friction, while inventory shows a threshold pattern in which sufficiently high friction favors tempered execution. Q2 then shows when that mechanism produces deployed misselection under one fixed interface.

Appendix robustness. Appendix checks surface robustness without changing the evidence hierarchy: event-micro includes alternate-threshold, log-loss, and hysteresis-interface checks; Traffic-Hourly includes a relative-rank variant; and expanded inventory and load-following remain fairness or appendix-only context. These checks are consistent with the main qualitative conclusion: under a fixed interface, a forecast-side winner can remain forecast-best while becoming recurrently deployed-suboptimal in moderate-to-high friction regimes.

Why forecast ordering need not be preserved. Proper forecast-side scores evaluate predictive quality, not the composed object obtained after forecasts are mapped to actions and executed through a frictional interface. If a forecaster's forecast-side advantage is smaller than the extra friction-induced loss created by its implied action path, the deployed ordering can reverse even under a fixed interface. This is why Q1 is useful as mechanism support and why Q2 is the actual benchmark-selection question.

3. Discussion, Related Work, and Limitations

Related work. Recent forecasting benchmarks span live AI-forecasting settings, broad time-series archives, and foundation-model evaluations (Karger et al., 2025; Yang et al., 2025; Aksu et al., 2024; Godahewa et al., 2021; Das et al., 2024; Williams et al., 2025). These efforts have made forecasting evaluation itself a central research problem, but they primarily compare models by forecast-side accuracy, calibration, or leaderboard performance. Our question is different: under one fixed forecast-to-decision interface, does the forecast-side winner remain the deployed-side winner? We therefore treat deployed-selection robustness as a reporting diagnostic for deployment-facing benchmarks rather than as a new forecasting model or optimizer, with variance-aware benchmark reporting providing the closest methodological motivation (Bouthillier et al., 2021).

This differs from proper scoring rules, decision-maker-perspective forecast evaluation, decision-focused learning, and predict-then-optimize, which either evaluate forecast quality/value or modify training/optimization objectives for downstream tasks (Gneiting and Raftery, 2007; Murphy, 1993; Raeth and Ludwig, 2025; Donti et al., 2017; Elmachtoub and Grigas, 2022; Mandi et al., 2024). Here, forecasters are fixed, and only the evaluation object changes.

Discussion and limitations. Our claim is evaluative and deliberately narrow. We do not claim that forecast-side metrics are invalid for prediction, and we do not seek a single reporting rule for all forecasting applications. We use low, moderate, and high only as shorthand for the lower, middle, and upper parts of each domain's tested friction grid; these labels are not intended as a universal calibration across domains. Event-micro provides the main forecasting-native evidence, Traffic-Hourly provides a second forecasting-native task family, and inventory provides operational corroboration, but this is still not a broad benchmark suite. Broader inventory sweeps are reported in the appendix, and load-following is secondary; the strongest main-text evidence comes from the moderate-to-high friction settings studied here. Our results therefore support a reporting recommendation: deployment-facing forecasting benchmarks should consider reporting fixed-interface deployed-selection robustness alongside standard forecast-side metrics.

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Minimum reporting standard for fixed-interface deployed-selection robustness. For each domain × interface × friction cell, report:

- domain/task, seed count, and evaluation split;
- fixed forecast-to-decision interface and interface parameters;
- friction grid, including a zero-friction anchor when applicable;
- forecast-side metric and tie-breaking rule;
- forecast-side winner and deployed-side winner;
- agreement rate between forecast-side and deployed-side selection;
- deployed-suboptimal share/count of the forecast-selected model;
- mean and median paired deployed gap with uncertainty intervals;
- whether the zero-friction row is exact, mixed, or not applicable.

A. Event-Micro Robustness Package

Because event-micro provides the paper's main forecasting-native Q2 evidence, this appendix records the first domain-specific robustness package for that benchmark. We construct a minimal forecasting-native binary-event setting evaluated over 100 seeds, in which each forecaster emits event probabilities, a fixed thresholding rule maps probabilities to actions, and switching friction penalizes action changes. The main text uses Brier as the canonical forecast-side criterion. This appendix records the full six-row Brier table plus an alternate-threshold check, a log-loss reranking check, and a second-interface hysteresis check.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap
0.00	Reactive sharp	Reactive sharp	0.62	0.003
0.05	Reactive sharp	Reactive sharp	0.60	0.002
0.10	Reactive sharp	Reactive sharp	0.58	0.002
0.25	Reactive sharp	Calibrated baseline	0.51	0.003
0.50	Reactive sharp	Calibrated baseline	0.31	0.011
1.00	Reactive sharp	Lagged smoother	0.01	0.057

Table A1. Forecast-side vs. deployed-side model selection in the full six-row event-micro benchmark summary. Under the canonical fixed-threshold interface, deployed-suboptimal counts rise from $38/100$ seeds at zero friction to $99/100$ at friction 1.00.

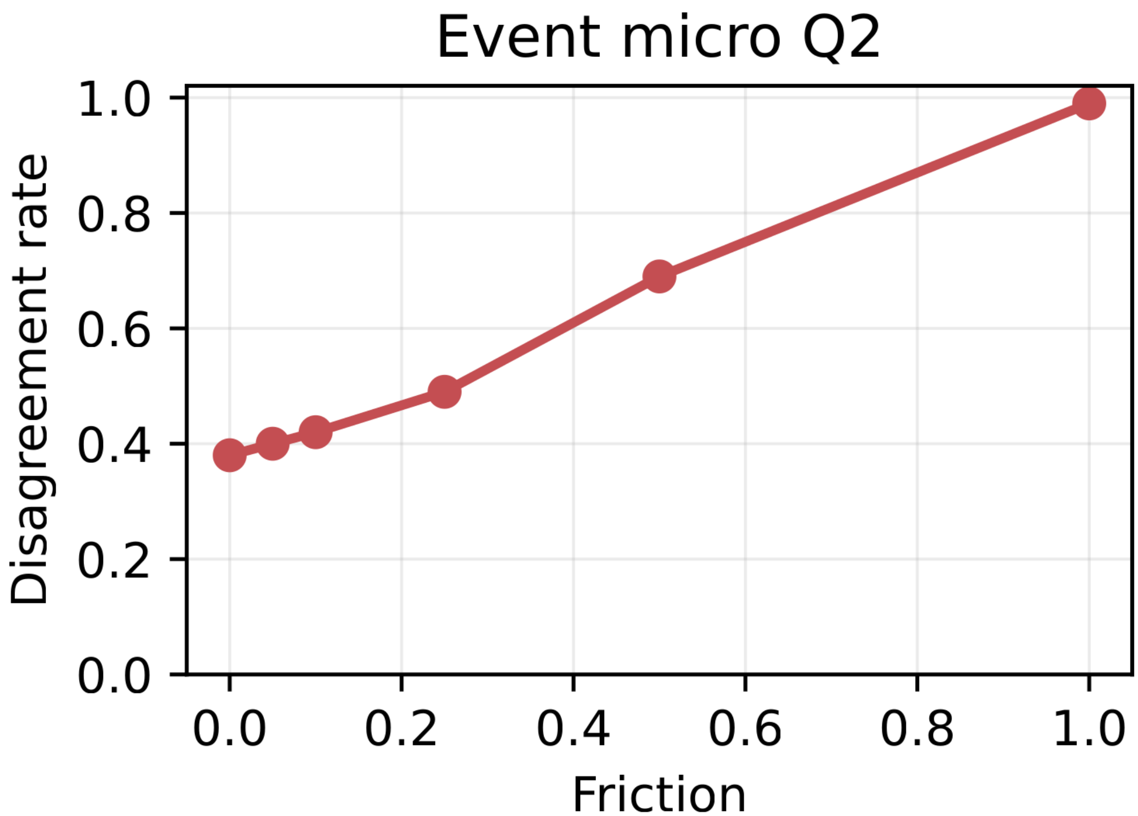

Figure A1. Ranking mismatch in a minimal event-forecasting micro-benchmark. Disagreement between forecast-side and deployed-side model selection is lowest at zero and very low friction and increases as switching frictions rise.

Threshold	Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
tau=0.55	0.00	Reactive sharp	Reactive sharp	0.62	0.003	0.000	38/100
tau=0.55	0.50	Reactive sharp	Calibrated baseline	0.31	0.011	0.008	69/100
tau=0.55	1.00	Reactive sharp	Lagged smoother	0.01	0.057	0.053	99/100
tau=0.50	0.00	Reactive sharp	Reactive sharp	0.56	0.002	0.000	44/100
tau=0.50	0.50	Reactive sharp	Calibrated baseline	0.23	0.014	0.011	77/100
tau=0.50	1.00	Reactive sharp	Lagged smoother	0.02	0.061	0.059	98/100

Table A2. Alternate-threshold robustness for the event-micro benchmark. The same qualitative winner drift and recurrent deployed-suboptimality pattern persists when the fixed threshold is changed from $\tau = 0.55$ to $\tau = 0.50$.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
0.00	Reactive sharp	Reactive sharp	0.62	0.002	0.000	38/100
0.50	Reactive sharp	Calibrated baseline	0.26	0.013	0.010	74/100
1.00	Reactive sharp	Lagged smoother	0.01	0.060	0.057	99/100

Table A3. Log-loss robustness for the event-micro benchmark at the canonical threshold. The same qualitative winner drift and recurrent deployed-suboptimality pattern persists when forecast-side ranking is recomputed with log loss instead of Brier.

Interface	Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
Hysteresis threshold	0.00	Reactive sharp	Reactive sharp	0.55	0.003	0.000	45/100
Hysteresis threshold	0.50	Reactive sharp	Calibrated baseline	0.45	0.005	0.001	55/100
Hysteresis threshold	1.00	Reactive sharp	Lagged smoother	0.18	0.022	0.020	82/100

Table A4. Second-interface robustness for the event-micro benchmark. Under a fixed hysteresis-threshold interface with $\delta = 0.05$, the same qualitative winner drift and deployed-suboptimality pattern remains visible, though the magnitudes are smaller than under the canonical fixed threshold.

Across this appendix robustness package, the same qualitative winner-drift and recurrent deployed-suboptimality pattern persists in a minimal forecasting-native event-micro benchmark evaluated over 100 seeds.

A.1. Interface-Aware Event-Micro Stats Addendum

This addendum records interface-specific recurrence and gap summaries for the hysteresis-threshold check without changing the main cross-domain recurrence package.

Interface	κ	Seeds	Agree.	Subopt. share/count	Subopt. 95% CI	CI type
Fixed threshold	0.50	100	0.31	0.69 (69/100)	[0.594, 0.772]	Wilson
Fixed threshold	1.00	100	0.01	0.99 (99/100)	[0.946, 0.998]	Wilson
Hysteresis threshold	0.50	100	0.45	0.55 (55/100)	[0.452, 0.644]	Wilson
Hysteresis threshold	1.00	100	0.18	0.82 (82/100)	[0.733, 0.883]	Wilson

Table A5. Interface-aware recurrence and share uncertainty for event-micro deployed-suboptimality. The canonical fixed threshold shows stronger recurrence, while hysteresis preserves the same direction with smaller magnitudes.

The aggregate recurrence table reports the seed-level pattern directly, replacing the dense seed-strip visualization.

Interface	Friction	Mean deployed gap	Mean gap 95% bootstrap CI	Median deployed gap	Median gap 95% bootstrap CI
Fixed threshold	0.50	0.011	[0.009, 0.014]	0.008	[0.004, 0.011]
Fixed threshold	1.00	0.057	[0.052, 0.063]	0.053	[0.048, 0.059]
Hysteresis threshold	0.50	0.005	[0.004, 0.006]	0.001	[0.000, 0.003]
Hysteresis threshold	1.00	0.022	[0.019, 0.026]	0.020	[0.015, 0.025]

Table A6. Interface-aware bootstrap intervals for the event-micro deployed gap. Positive intervals remain visible under both interfaces, with smaller gaps under hysteresis.

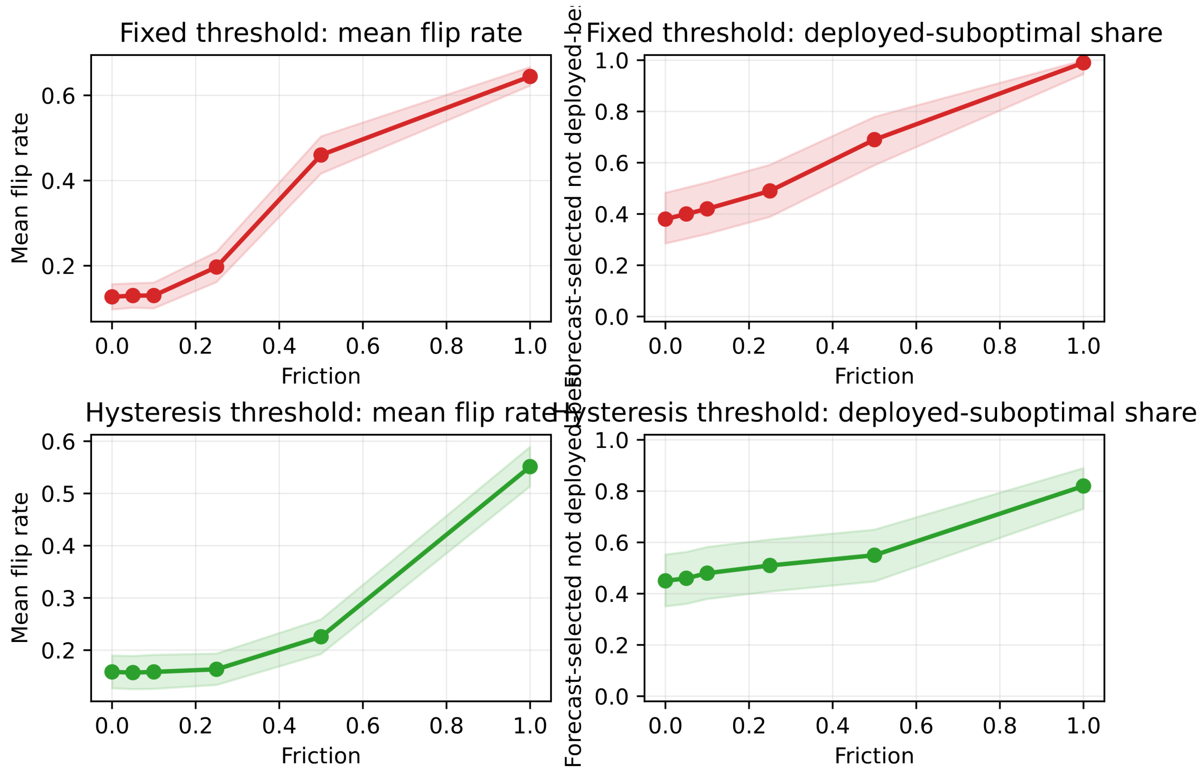

Figure A2. Event-micro robustness across interfaces. Fixed-threshold and hysteresis interfaces show the same direction: disagreement and deployed-suboptimal share rise with friction, while hysteresis attenuates the effect.

B. Traffic-Hourly Forecasting-Native Checks

B.1. Fixed-Budget Top-k Alert

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
0.00	Reactive short	Reactive short	1.00	0.000	0.000	0/100
0.50	Reactive short	Lagged smoother	0.00	7.016	6.977	100/100
1.00	Reactive short	Lagged smoother	0.00	24.651	24.473	100/100

Table A7. Traffic-Hourly Top-k forecasting-native Q2 check under a fixed-budget interface. Agreement is strongest at zero friction and weakens at higher friction, where the deployed winner shifts away from the forecast-side winner.

In the selected Top-k setting, Reactive short is the forecast-side winner and the deployed winner at zero friction. At frictions 0.5 and 1.0, the deployed winner shifts to Lagged smoother even though Reactive short remains Brier-best and continues to win on the forecast side. The qualitative reading is the same as in the main text: under a fixed interface, the more reactive forecast-optimal family need not remain deployment-optimal once switching is penalized.

B.2. Relative-Rank Traffic Variant

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
0.00	Reactive short	Reactive short	1.00	0.000	0.000	0/100
0.50	Reactive short	Calibrated baseline	0.00	10.103	10.122	100/100
1.00	Reactive short	Calibrated baseline	0.00	25.205	25.231	100/100

Table A8. Traffic-Hourly relative-rank appendix variant. The same qualitative separation between forecast-side and deployed-side selection appears under fixed-budget set action.

The relative-rank target shows the same qualitative separation under a different forecasting-native label construction. Reactive short remains the forecast-side winner, while Calibrated baseline becomes the deployed winner once switching costs are introduced, so we report this variant as an appendix forecasting-native check. The alternate $m = 0.15$ variant is also strong in the full report but is omitted from the compact table to keep the appendix narrow.

C. Inventory Operational Corroboration and Fairness Context

C.1. Five-Family Inventory Corroboration

This appendix records the full five-family inventory summary that underlies the bounded main-text inventory evidence. The mixed zero and low-friction rows are reported here, while the main text uses only the moderate-to-high friction rows to avoid overstating the zero-friction story.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Deployed-suboptimal seeds / total
0.00	Small MLP	Small MLP	0.77	0.025	23/100
0.25	Small MLP	Small MLP	0.76	0.039	24/100
0.50	Small MLP	Moving average (7)	0.26	0.240	74/100
1.00	Small MLP	Moving average (7)	0.01	1.029	99/100

Table A9. Inventory corroboration under a fixed replenishment interface. The zero/low-friction regime is mixed, while moderate-to-high friction shows recurrent deployed-side misselection.

C.2. Expanded-Family Inventory Fairness Context

The stronger-baseline sweep is included as a fairness check for the inventory corroboration. It contextualizes the inventory corroboration; the main claim is supported by the fixed-interface selection comparison in the main text.

Domain	Role	Status	Seeds	Zero flip	Zero rho	Best +fric.	Best +flip	Strongest pair	Flip share	Note
Synthetic	required	clears stricter gate	20	0.000	1.000	0.2500	0.500	Linear AR / Reactive heuristic	1.000	passed stricter gate
Inventory	required	mixed zero row	100	0.105	0.870	1.0000	0.467	Moving average (7) / Naive persistence	1.000	does not clear stricter gate

Table A10. Expanded same-interface inventory sweep. The table records broader-family status checks for inventory robustness.

Detailed tuning grids and representative selection are summarized compactly; the robustness table reports the resulting fixed-interface comparison.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Deployed-suboptimal seeds / total
0.00	Small MLP	Gradient-boosted trees	0.60	0.043	4/10
0.25	Small MLP	Gradient-boosted trees	0.60	0.059	4/10
0.50	Small MLP	Gradient-boosted trees	0.50	0.128	5/10
1.00	Small MLP	Moving average (7)	0.10	0.684	9/10

Table A11. Expanded-family inventory Q2 robustness summary for the broader comparison set.

The mixed zero/low-friction rows do not overturn the narrower main claim from the five-family inventory comparison.

D. Load-Following Appendix-Only Secondary Check

Load-following is a secondary operational check. It tests whether forecast-side and deployed-side selection can diverge outside the event-micro and Traffic-Hourly settings.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
0.00	Linear AR	Small MLP	0.70	0.000	0.000	3/10
0.25	Linear AR	Small MLP	0.30	0.001	0.000	7/10
0.50	Linear AR	Small MLP	0.30	0.001	0.001	7/10
1.00	Linear AR	Small MLP	0.20	0.002	0.002	8/10

Table A12. Load-following secondary check. Near-zero friction remains mixed, while positive-friction regimes can favor a different deployed winner.

The result is consistent with the narrower reading used throughout the paper: deployed-side selection can differ from forecast-side selection under friction, but this domain is not used as primary evidence.

E. Q1 Mechanism Support

Q1 serves only as mechanism support for why Q2 divergence can arise. Holding the proposed action path fixed, synthetic and inventory show how a fixed interface can separate proposed and realized actions once friction is introduced.

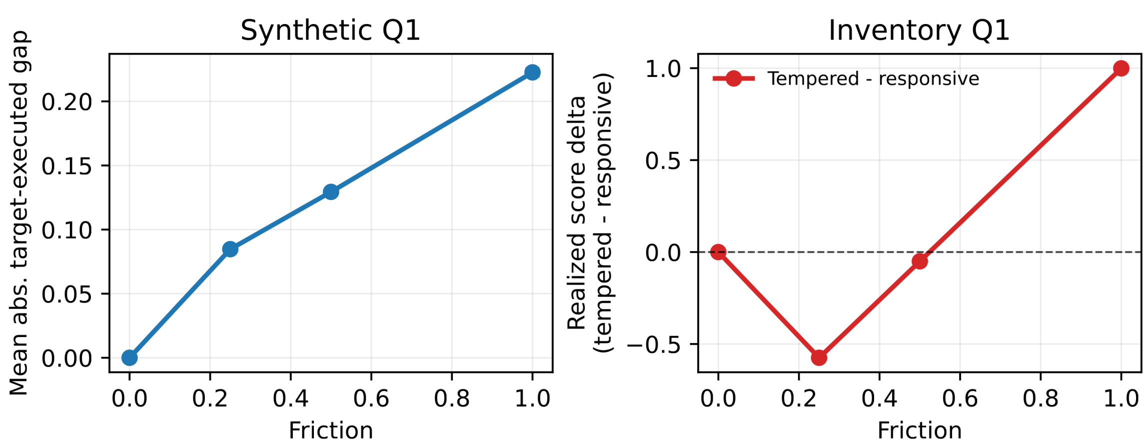

Figure A3. Q1 mechanism support. Even with the same proposed actions, a fixed frictional interface can alter realized actions and realized score; this explains how Q2 ranking divergence can arise.

F. Uncertainty Reporting for Seed-Level Recurrence and Deployed Gaps

This appendix reports uncertainty for the seed-level recurrence and paired deployed-gap summaries used in the main report card. We emphasize intervals and paired shortfalls rather than hypothesis-test significance. For binary share summaries, rows with fewer than 30 matched seeds use Clopper–Pearson exact intervals; rows with at least 30 matched seeds use Wilson intervals. For continuous deployed gaps, we use 10,000 paired seed bootstrap resamples. We report these intervals as descriptive uncertainty summaries for the reporting diagnostic, not as hypothesis-test evidence for a universal effect.

Domain	Interface	κ	Seeds	Agree.	Subopt. share/count	Subopt. 95% CI	CI type
Event-micro	threshold $\tau = 0.55$	0.50	100	0.31	0.69 (69/100)	[0.594, 0.772]	Wilson
Event-micro	threshold $\tau = 0.55$	1.00	100	0.01	0.99 (99/100)	[0.946, 0.998]	Wilson
Traffic Top-k	budget $k = 249$	0.50	100	0.00	1.00 (100/100)	[0.963, 1.000]	Wilson
Traffic Top-k	budget $k = 249$	1.00	100	0.00	1.00 (100/100)	[0.963, 1.000]	Wilson
Inventory	replenishment	0.50	100	0.26	0.74 (74/100)	[0.646, 0.816]	Wilson
Inventory	replenishment	1.00	100	0.01	0.99 (99/100)	[0.946, 0.998]	Wilson
Load-following	secondary	0.25	10	0.30	0.70 (7/10)	[0.348, 0.933]	Clopper–Pearson
Load-following	secondary	0.50	10	0.30	0.70 (7/10)	[0.348, 0.933]	Clopper–Pearson
Load-following	secondary	1.00	10	0.20	0.80 (8/10)	[0.444, 0.975]	Clopper–Pearson

Table A13. Seed-level recurrence and share uncertainty for deployed-suboptimality. Agreement rate is the fraction of matched seeds for which the forecast-side selected model is also the deployed-side best model; deployed-suboptimal share is the complementary fraction. We report Clopper–Pearson exact intervals for rows with $n < 30$ and Wilson intervals otherwise.

Aggregate recurrence intervals and uncertainty bands summarize the seed-level pattern without the dense seed-strip visualization.

Domain	Friction	Mean deployed gap	Mean gap 95% bootstrap CI	Median deployed gap	Median gap 95% bootstrap CI
Event micro	0.50	0.011	[0.009, 0.014]	0.008	[0.004, 0.011]
Event micro	1.00	0.057	[0.052, 0.063]	0.053	[0.048, 0.059]
Traffic-Hourly Top-k	0.50	7.016	[6.756, 7.283]	6.977	[6.706, 7.281]
Traffic-Hourly Top-k	1.00	24.651	[24.226, 25.079]	24.473	[24.001, 25.051]
Inventory	0.50	0.240	[0.195, 0.286]	0.202	[0.154, 0.256]
Inventory	1.00	1.029	[0.946, 1.111]	1.045	[0.926, 1.167]
Load-following	0.25	0.001	[0.000, 0.001]	0.000	[0.000, 0.001]
Load-following	0.50	0.001	[0.000, 0.002]	0.001	[0.000, 0.002]
Load-following	1.00	0.002	[0.001, 0.005]	0.002	[0.000, 0.003]

Table A14. Paired-bootstrap intervals for deployed-gap rows. Deployed gap is the paired deployed-utility shortfall $U_{\text{best}} - U_{\text{forecast-selected}}$ computed within each matched seed. Intervals use 10,000 paired seed bootstrap resamples. Event-micro and Traffic-Hourly provide the main forecasting-native Q2 gap evidence; inventory is operational corroboration; load-following remains appendix-only directional context.

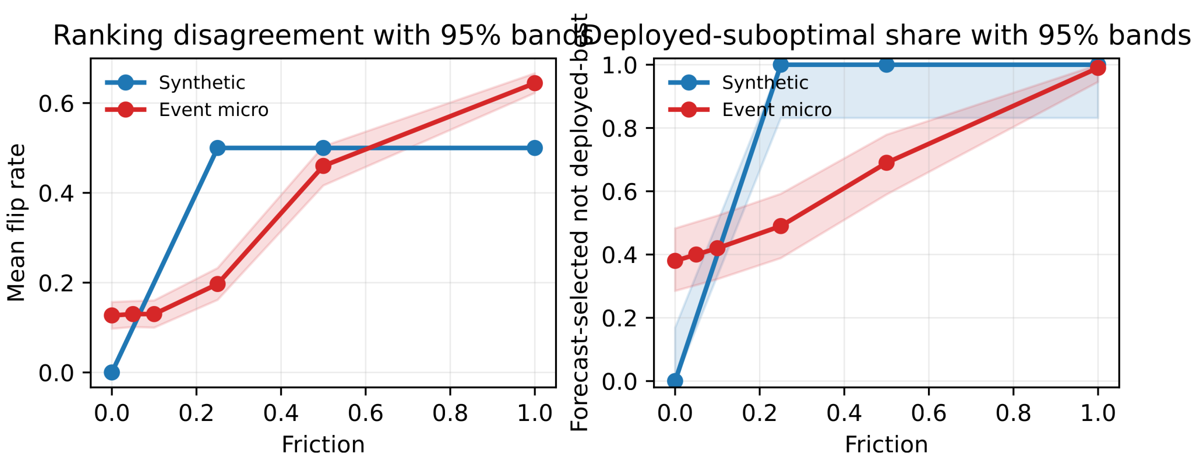

Figure A4. Q2 uncertainty bands for synthetic and event-micro. Panels show mean flip-rate bands and share-uncertainty intervals for deployed-suboptimality, using the interval convention described in Appendix F.